Pratham Modi

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AI Engineer | Python, Generative AI, Data Pipelines, Cloud Infrastructure

Summary

AI Software Engineer with 2+ years of production experience building high-throughput data engineering pipelines, GenAI agent workflows, and scalable backend services on AWS. Engineered real-time WebSocket communication infrastructure and Kafka event-streaming pipelines handling 75K+ daily events with sub-50ms latency at C3iHub (IIT Kanpur), orchestrated multi-model GenAI workflows and RAG systems, and optimized state tracking across PostgreSQL and Redis. Hands-on expertise in Python, FastAPI, Docker, and distributed systems.

Technical Skills

Languages: Python (Asyncio, Profiling), Node.js, TypeScript, SQL, Go, Java, C++
AI & Machine Learning: Generative AI (GenAI), LLM Orchestration, Prompt Engineering, RAG Pipelines, Vector Databases, Computer Vision Basics
Backend & Data Engineering: Python (FastAPI), RESTful APIs, Microservices, Data Pipelines, Event-Driven Architecture
Data & Caching: Redis, Redis Streams, PostgreSQL, Apache Kafka, Distributed Transactions
Cloud & DevOps: AWS (EC2, S3, EKS), Docker, Kubernetes, CI/CD, OpenTelemetry, Prometheus, Grafana

Experience

C3iHub, IIT Kanpur Jul 2025 – Present
Software Development Engineer (AI Systems & Infrastructure) Bengaluru / Kanpur, India

- Architected core platform backend and real-time streaming infrastructure serving 50K+ active users across distributed microservices, ensuring 99.9% uptime via automated health reconciliation.
- Built real-time interactive audio and message streaming modules using WebSockets and async Python workers, achieving sub-100ms transport latencies under concurrent user sessions.
- Engineered event-driven streaming pipeline handling 75K+ daily events with sub-50ms latency using Apache Kafka and Redis Streams – implemented consumer-group partition balancing and idempotent consumer processing.
- Optimized data retrieval queries across PostgreSQL and Redis, slashing p99 state latency from 450ms to 135ms.
- Automated container build, vulnerability scanning, and deployment pipelines on AWS using GitHub Actions, cutting staging deploy cycles by 55%.

Playpower Labs Jan 2024 – Jun 2025
Software Development Engineer (Backend & AI Services) Bengaluru, India

- Built high-performance async microservices in Python (FastAPI) and Node.js handling 500+ persistent concurrent connections with p95 response latencies under 85ms.
- Orchestrated Generative AI pipelines and automated tool execution for intelligent data extraction and reasoning workflows.
- Implemented distributed caching and query optimization across PostgreSQL and Redis, improving database throughput by 40% under burst traffic workloads.
- Integrated OpenTelemetry distributed tracing across 15+ backend services, instrumenting unified spans that reduced cross-service incident resolution time by 35%.

Projects

Autonomous Agent Orchestrator & Data Pipeline | *Python, FastAPI, Redis, PostgreSQL, Docker, AWS*

- Engineered end-to-end multi-agent evaluation pipeline with automated verification, LLM tool-calling, and trace logging for deterministic execution benchmarks.
- Implemented session context and persistent state management using Redis Streams and PostgreSQL, enabling seamless multi-turn conversation recovery.

Education

Pandit Deendayal Energy University 2021 – 2025
Bachelor of Technology in Computer Science and Engineering CGPA: 9.20 / 10.0